

Notes

Organism VS Species

Organism

Organism refers to a living things that :

1. Has an organised structure
2. react to any scenario
3. grow over time
4. adapts to environments
5. reproduce
6. maintain homeostasis

Species

Species is a group of organism that has similar genetical and physical characteristics. All organism belong to a particular species.

What is bio_informatics ?



Bioinformatics is an interdisciplinary field of science in which biology, statistics/mathematics, computer science and information technology merge together to form a single discipline in order to analyze biological information using computers and statistical techniques.

Goals of bio_informatics

1. To construct and manage biological database.
2. To develop algorithm for different types of sequence analysis.

3. To develop softwares.

Purpose of bio_informatics

- The work of bioinformatics is in the processing of large amount of data.
- The purpose of bioinformatics is to make sense of processes taking place inside organism or living systems.

Use of bio_informatics

Bioinformatics is used in various fields, especially in biomedicine where it has various applications.

Drug discovery: By the use of structure-based drug design with Bioinformatics, scientists can come up with effective drugs for various illnesses and acute diseases.

Personalized medicine: By assessing the genetic structure of the patient and their respective medical history, personalized medicine can be curated which would prove to be much more effective.

Preventive medicines: preventive medicines can prevent a chain of transmission of diseases or mitigate any acute disease before its onset.

Gene therapy: Gene therapy is the process where defective genes are replaced by new ones in the gene structure of a living organism. Since each organisms gene structure is very different, huge data may be involved to produce an accurate replacement.

What is a Ribosome?



A **ribosome** is a complex molecular machine found in both prokaryotic and eukaryotic cells. It is responsible for **protein synthesis**, translating genetic information from messenger RNA (**mRNA**) into proteins. Ribosomes are composed of **ribosomal RNA (rRNA)** and **proteins**.

Structure of Ribosome

Ribosomes consist of two subunits:

◆ Prokaryotic Ribosome (70S)

- **Small Subunit (30S)** – Contains **16S rRNA** and proteins.
- **Large Subunit (50S)** – Contains **23S rRNA**, **5S rRNA**, and proteins.

◆ Eukaryotic Ribosome (80S)

- **Small Subunit (40S)** – Contains **18S rRNA** and proteins.
- **Large Subunit (60S)** – Contains **28S rRNA**, **5.8S rRNA**, **5S rRNA**, and proteins.

📌 The "S" (Svedberg unit) represents sedimentation rate, not size.

Functional Sites in the Ribosome

1. **A-site (Aminoacyl site)** – Entry site for tRNA carrying new amino acids.
2. **P-site (Peptidyl site)** – Holds the tRNA with the growing peptide chain.
3. **E-site (Exit site)** – Releases tRNA after amino acid transfer.

Ribosome's Role in DNA Replication, Transcription, and Translation

1. Replication (DNA Synthesis)

- Ribosomes are **NOT** directly involved in replication.

- Replication is carried out by **DNA polymerase** in the nucleus (eukaryotes) or nucleoid region (prokaryotes).
- However, proteins required for replication (e.g., helicase, primase) are made by ribosomes.

2. Transcription (RNA Synthesis)

- *Ribosomes do not play a direct role in transcription.*
- Transcription is performed by **RNA polymerase**, which synthesizes mRNA from DNA.
- Ribosomes are involved indirectly, as they produce the proteins (e.g., transcription factors) needed for this process.

3. Translation (Protein Synthesis)

Ribosomes are the **central players in translation**, where mRNA is decoded into a protein.

Summary

- **Ribosomes** are essential for **protein synthesis** but do not directly participate in DNA replication or transcription.
- They are composed of two subunits and have **A, P, and E sites** for translation.
- In **translation**, they decode mRNA and assemble proteins using **tRNA and amino acids**.

Definitions

1. Replication



Replication is the process of copying **DNA** to produce an identical **DNA molecule**. It occurs in the **nucleus (eukaryotes)** or **nucleoid region (prokaryotes)** and is carried out by **DNA polymerase**, ensuring genetic information is accurately passed to daughter cells.

2. Transcription



Transcription is the process of synthesizing **RNA** from a **DNA template**. It is carried out by **RNA polymerase** and results in the formation of **mRNA, tRNA, or rRNA**. This process occurs in the **nucleus (eukaryotes)** or **cytoplasm (prokaryotes)**.

3. Translation



Translation is the process of synthesizing **proteins** using the **mRNA sequence** as a template. It occurs in the **ribosome**, where **tRNA** brings amino acids to form a polypeptide chain based on the codon sequence in mRNA.

Explanation of Replication, Transcription, and Translation

The flow of genetic information follows the **Central Dogma of Molecular Biology**, which describes how DNA is copied, transcribed into RNA, and translated into proteins. Let's go step by step.

DNA Replication (DNA → DNA)

Definition:

Replication is the process of copying **DNA** to produce an identical DNA molecule before cell division.

Where It Occurs:

- **Eukaryotes:** Nucleus
- **Prokaryotes:** Cytoplasm (nucleoid region)

Steps of DNA Replication

1. Initiation

- The enzyme **helicase** unwinds(শান্ত করা) the double helix by breaking **hydrogen bonds** between base pairs.
- **Single-strand binding proteins (SSBPs)** stabilize the unwound DNA.
- **Topoisomerase** prevents supercoiling.

2. Elongation

- **Primase** synthesizes RNA primers to provide a starting point.
- **DNA polymerase III** adds nucleotides to the **leading strand** continuously (5' → 3' direction).
- **Lagging strand** is synthesized in short fragments (**Okazaki fragments**) using multiple primers.
- **DNA polymerase I** replaces RNA primers with DNA.
- **DNA ligase** joins Okazaki fragments, forming a continuous strand.

3. Termination

- Replication ends when two replication forks meet or reach **telomeres** (in eukaryotes).
- **Telomerase** extends the ends of chromosomes (only in eukaryotes).

 **Final Product:** Two identical DNA molecules.

Transcription (DNA → RNA)

Definition:

Transcription is the process of synthesizing **RNA** from a DNA template.

Where It Occurs:

- **Eukaryotes:** Nucleus

- **Prokaryotes:** Cytoplasm

Steps of Transcription

1. Initiation

- **RNA polymerase** binds(আবদ্ধ করে) to the **promoter region** of DNA (e.g., TATA box in eukaryotes).
- DNA strands unwind, and one strand (**template strand**) is used to **synthesize RNA**.

2. Elongation

- RNA polymerase moves along the DNA template, adding complementary **ribonucleotides** (A → U, T → A, G → C, C → G).
- The RNA strand grows in the **5' → 3' direction**.

3. Termination

- RNA polymerase reaches a **termination sequence** (e.g., hairpin loop in prokaryotes).
- The newly synthesized **pre-mRNA** (in eukaryotes) or **mRNA** (in prokaryotes) is released.

4. Post-Transcriptional Modifications (Eukaryotes Only)

- **5' Cap** (modified guanine) is added for stability.
- **Poly-A Tail** (multiple adenines) is added to the 3' end.
- **Splicing** removes introns and joins exons.

 **Final Product:** Mature mRNA (ready for translation).

Translation (mRNA → Protein)

Definition

Translation is the process of synthesizing **proteins** using an mRNA template.

Where It Occurs

- **Eukaryotes & Prokaryotes:** Ribosomes in the cytoplasm.

Steps of Translation

1. Initiation

- The **small ribosomal subunit** binds to the mRNA at the start codon (**AUG**).
- The **initiator tRNA** (carrying methionine) binds to the P-site.
- The **large ribosomal subunit** joins, forming the functional ribosome.

2. Elongation

- A new tRNA carrying an amino acid enters the A-site.
- A **peptide bond** forms between amino acids in the P-site and A-site.
- The ribosome moves along mRNA (translocation), shifting tRNA from A → P → E.

3. Termination

- When a **stop codon (UAA, UAG, or UGA)** reaches the A-site, a **release factor** binds.
- The newly synthesized **polypeptide chain** is released.
- The ribosome dissociates into its subunits.

 **Final Product:** A functional protein.

Summary of the Three Processes

Process	Purpose	Enzyme Involved	Product
Replication	Copies DNA before division	DNA polymerase	DNA
Transcription	Converts DNA to RNA	RNA polymerase	mRNA
Translation	Converts mRNA to protein	Ribosome & tRNA	Protein

Describe the Central Dogma of Biology

The **Central Dogma of Molecular Biology** describes the flow of genetic information in a biological system. It was first proposed by **Francis Crick** in 1958 and states that **DNA directs the synthesis of RNA, which then directs the synthesis of proteins**.

Steps of the Central Dogma

1. Replication (DNA → DNA)

- DNA is copied to produce an identical DNA molecule.
- Enzyme: **DNA Polymerase**
- Occurs in: **Nucleus (Eukaryotes), Cytoplasm (Prokaryotes)**

2. **Transcription (DNA → RNA)**

- DNA is transcribed into messenger RNA (**mRNA**).
- Enzyme: **RNA Polymerase**
- Occurs in: **Nucleus (Eukaryotes), Cytoplasm (Prokaryotes)**

3. **Translation (RNA → Protein)**

- mRNA is translated into a protein by ribosomes.
- Enzyme: **Ribosome + tRNA**
- Occurs in: **Cytoplasm (Both Prokaryotes & Eukaryotes)**

Diagram Representation

Here's a simple way to visualize the Central Dogma:

DNA → (Transcription) → RNA → (Translation) → Protein

Exceptions to the Central Dogma

1. **Reverse Transcription (RNA → DNA)**

- Found in **retroviruses** like HIV.
- Enzyme: **Reverse Transcriptase**

2. **Non-Coding RNA (ncRNA)**

- Some RNAs (e.g., **tRNA, rRNA, miRNA**) do not code for proteins but perform cellular functions.

TATA Box: Definition and Function

The **TATA box** is a **DNA sequence** found in the promoter region of many **genes** in **eukaryotes** and **archaea**. It is named after its conserved "**TATAAA**" sequence and is located about **25–35 base pairs upstream** of the transcription start site (TSS).

Function of the TATA Box

1. Initiates Transcription

- The TATA box helps recruit the **RNA polymerase II** enzyme to begin **transcription**.

2. Binds Transcription Factors

- A protein called **TATA-binding protein (TBP)** binds to the TATA box.
- TBP is part of a larger complex called **TFIID**, which helps position RNA polymerase II at the correct site.

3. Regulates Gene Expression

- The TATA box ensures **precise** and **efficient** transcription initiation for genes that require **high regulation**, such as developmental or stress-response genes.

Location and Presence

- Found in **eukaryotic promoters**, especially in genes with high transcription rates.
- Present in some **archaeal** genes but rare in **prokaryotic** genes.
- Not all eukaryotic genes have a TATA box; many use alternative promoter elements like **GC-rich boxes** or **enhancers**.

Diagram of the TATA Box:

5' - TATAAA - 3' (Promoter region)



(TBP binds here)



RNA Polymerase II binds



Transcription begins

Analogy for Better Understanding

Think of the **TATA box** as a **starting signal** or a **marker** on a race track. It tells the **RNA polymerase** where to start “running” (transcribing DNA into RNA).

Working Principle and Functions of Key Enzymes in DNA Replication, Transcription, and Translation

DNA replication, transcription, and translation involve various **enzymes** that facilitate the process efficiently. Below is a breakdown of key enzymes, their functions, and how they work.

DNA Replication Enzymes (DNA → DNA)

DNA replication is the process by which DNA makes an identical copy of itself. It occurs in the **S phase** of the cell cycle.

Key Enzymes & Their Functions:

Enzyme	Function	Working Principle
Helicase	Unwinds the DNA double helix.	Breaks hydrogen bonds between base pairs, creating a replication fork.
Gyrase (Topoisomerase II)	Relieves supercoiling stress.	Cuts and reseals the DNA strand to prevent tangling.
Single-Strand Binding Proteins (SSBPs)	Stabilizes single-stranded DNA.	Prevents the DNA strands from reannealing.
Primase	Synthesizes RNA primers.	Provides a short RNA primer for DNA polymerase to begin synthesis.
DNA Polymerase III	Main DNA synthesizing enzyme.	Adds nucleotides to the growing strand in the 5' → 3' direction.
DNA Polymerase I	Replaces RNA primers with DNA.	Removes RNA primers and fills the gaps with DNA nucleotides.
DNA Ligase	Seals gaps between Okazaki fragments.	Forms phosphodiester bonds , creating a continuous DNA strand.

Working Principle

1. **Helicase unwinds** the DNA, forming a **replication fork**.
2. **Gyrase relieves tension** by cutting and rejoining DNA strands.
3. **SSBPs stabilize** the unwound strands.
4. **Primase lays down RNA primers** to provide a starting point.
5. **DNA Polymerase III extends the strand**, synthesizing **new DNA**.
6. **DNA Polymerase I replaces RNA primers** with DNA.
7. **DNA Ligase seals** the nicks between Okazaki fragments.

Transcription Enzymes (DNA → RNA)

Transcription is the process where DNA is converted into **messenger RNA (mRNA)**.

Key Enzymes & Their Functions:

Enzyme	Function	Working Principle
RNA Polymerase	Synthesizes mRNA from DNA.	Binds to the promoter region and adds RNA nucleotides complementary to the DNA template.
Helicase (in transcription)	Unwinds DNA for transcription.	Separates DNA strands so RNA polymerase can access the template.
TATA-binding protein (TBP)	Recognizes and binds the TATA box.	Helps RNA polymerase bind to the promoter.
Transcription Factors	Regulate transcription initiation.	Help recruit RNA polymerase to the correct start site.

Working Principle

1. **RNA Polymerase binds to the promoter** with the help of **TBP** and transcription factors.
2. **Helicase unwinds DNA**, exposing the template strand.
3. **RNA Polymerase adds complementary RNA nucleotides** (A, U, G, C) in the **5' → 3' direction**.
4. **Transcription continues until a termination signal** is reached.

5. **The mRNA strand detaches** and undergoes processing (splicing, capping, polyadenylation).

Translation Enzymes (RNA → Protein)

Translation is the process where **mRNA is decoded** into a **polypeptide (protein)** at the ribosome.

Key Enzymes & Their Functions

Enzyme / Factor	Function	Working Principle
Ribosome (rRNA + Proteins)	Catalyzes protein synthesis.	Provides a site for mRNA-tRNA interactions.
tRNA (Transfer RNA)	Brings amino acids to ribosome.	Matches mRNA codons with its anticodon and carries the correct amino acid.
Aminoacyl-tRNA Synthetase	Attaches amino acids to tRNA.	Links specific amino acids to corresponding tRNA molecules.
Peptidyl Transferase (Part of Ribosome)	Forms peptide bonds.	Catalyzes the formation of peptide bonds between amino acids.
Initiation Factors (IFs)	Help ribosome assemble on mRNA.	Guide the small ribosomal subunit to the mRNA start codon (AUG).
Elongation Factors (EFs)	Assist in adding amino acids.	Ensure the correct tRNA enters the ribosome.
Release Factors (RFs)	Terminate translation.	Recognize stop codons and release the completed polypeptide.

Working Principle

1. **Initiation:** The ribosome assembles at the mRNA **start codon (AUG)** with the help of **Initiation Factors**.
2. **Elongation:**
 - a. **tRNA molecules bring amino acids** to the ribosome.
 - b. **Peptidyl Transferase forms peptide bonds** between amino acids.
 - c. The ribosome moves along the mRNA, adding amino acids to the growing protein chain.
3. **Termination**

- a. A **stop codon (UAA, UGA, or UAG)** is reached.
- b. **Release Factors** help disassemble the ribosome, releasing the protein.

Summary Table of Enzymes in Replication, Transcription, and Translation:

Process	Enzyme	Function
Replication	Helicase	Unzips DNA strands
	Gyrase	Relieves supercoiling
	Primase	Lays RNA primers
	DNA Polymerase III	Synthesizes new DNA
	DNA Polymerase I	Replaces RNA primers
	Ligase	Joins Okazaki fragments
Transcription	RNA Polymerase	Synthesizes mRNA from DNA
	Helicase	Unwinds DNA for transcription
	TATA-binding protein	Helps RNA Polymerase bind to the promoter
	Transcription Factors	Regulate transcription initiation
Translation	Ribosome	Facilitates protein synthesis
	tRNA	Delivers amino acids
	Aminoacyl-tRNA Synthetase	Attaches amino acids to tRNA
	Peptidyl Transferase	Forms peptide bonds
	Initiation Factors	Assemble the ribosome
	Elongation Factors	Assist in amino acid addition
	Release Factors	Terminate translation

Final Notes

- **Replication** ensures DNA is copied before cell division.
- **Transcription** converts genetic information into mRNA.
- **Translation** produces proteins essential for life.

Replication Fork: Definition & Function

Definition

A **replication fork** is the Y-shaped structure that forms when **DNA helicase** unwinds the double-stranded DNA during **DNA replication**. It serves as the active site where new DNA strands are synthesized.

Structure of the Replication Fork

At the replication fork, the DNA strands separate into:

1. **Leading Strand** → Synthesized continuously in the **5' → 3' direction**.
2. **Lagging Strand** → Synthesized **discontinuously** in short segments called **Okazaki fragments** (since DNA polymerase can only add nucleotides in the 5' → 3' direction).

Key Components at the Replication Fork

Component	Function
Helicase	Unwinds DNA by breaking hydrogen bonds.
Single-Strand Binding Proteins (SSBs)	Prevents reannealing of separated strands.
DNA Gyrase (Topoisomerase)	Relieves supercoiling stress ahead of the fork.
Primase	Lays down RNA primers for DNA polymerase.
DNA Polymerase III	Extends the leading strand and forms Okazaki fragments on the lagging strand.
DNA Polymerase I	Replaces RNA primers with DNA nucleotides.
DNA Ligase	Joins Okazaki fragments by forming phosphodiester bonds.

Working Mechanism of the Replication Fork

1. **Helicase unwinds** the DNA, creating two single-stranded templates.
2. **SSBs bind** to prevent reannealing of DNA strands.
3. **Primase lays primers** on both the leading and lagging strands.
4. **DNA Polymerase III synthesizes DNA:**

- **Leading strand:** Continuously in **5' → 3'** direction.
 - **Lagging strand:** Discontinuously as **Okazaki fragments**.
5. **DNA Polymerase I replaces RNA primers** with DNA.
 6. **DNA Ligase seals gaps** between Okazaki fragments.
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Why is the Replication Fork Important?

- Ensures accurate and fast DNA replication.
- Prevents errors that could lead to mutations.
- Supports **semi-conservative replication** (each new DNA has one old strand and one new strand).

Okazaki Fragments: Definition & Function

Definition

Okazaki fragments are short sequences of **newly synthesized DNA** that are formed on the **lagging strand** during DNA replication. These fragments are later joined together by **DNA ligase** to create a continuous strand.

Why Are Okazaki Fragments Needed?

DNA polymerase can only synthesize DNA in the **5' → 3' direction**. Since the two strands of DNA are **antiparallel**, replication occurs differently:

- The **leading strand** is synthesized **continuously** in the **5' → 3' direction**.
 - The **lagging strand** runs in the **3' → 5' direction**, so it is synthesized **discontinuously** in short segments (Okazaki fragments).
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Process of Okazaki Fragment Formation

1. **Helicase unwinds DNA**, forming a replication fork.
2. **Primase lays down RNA primers** at intervals on the lagging strand.
3. **DNA Polymerase III extends** each RNA primer by adding nucleotides in the **5' → 3' direction**, forming Okazaki fragments.
4. **DNA Polymerase I removes the RNA primers** and replaces them with DNA.

5. **DNA Ligase seals the gaps** between fragments by forming **phosphodiester bonds**, creating a continuous strand.

Key Enzymes Involved

Enzyme	Function
Primase	Lays down RNA primers for DNA polymerase.
DNA Polymerase III	Synthesizes Okazaki fragments by adding nucleotides.
DNA Polymerase I	Replaces RNA primers with DNA.
DNA Ligase	Joins Okazaki fragments into a continuous strand.

Importance of Okazaki Fragments

- Ensure that the entire **lagging strand** is accurately copied.
- Maintain the **semi-conservative nature** of DNA replication.
- Help prevent errors and maintain genetic integrity.

DNA Replication Models

Before the correct mechanism of DNA replication was confirmed, scientists proposed three possible models:

1. **Conservative Model**
2. **Semi-Conservative Model** (Correct Model)
3. **Dispersive Model**

Conservative Model (Incorrect)

- The **entire DNA molecule remains intact**, and a completely **new DNA molecule** is synthesized.
- After replication, one DNA molecule is **100% old**, and the other is **100% newly synthesized**.
- **Rejected** by the Meselson-Stahl experiment.

Semi-Conservative Model (Correct)

- Each **new DNA molecule** consists of **one old (parental) strand** and **one newly synthesized strand**.

- **Supported by the Meselson-Stahl experiment** using **nitrogen isotopes** (N^{15} and N^{14}).
 - Ensures **genetic stability** across generations.
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Dispersive Model (Incorrect)

- DNA strands **break into small fragments**, and new DNA is synthesized in a **mixed pattern** of old and new segments.
 - After replication, each DNA strand contains **randomly mixed old and new fragments**.
 - **Rejected** by experimental evidence.
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Final Conclusion

The **semi-conservative model** is the correct DNA replication mechanism, as **each new DNA molecule retains half of the original strand** and **builds a new complementary strand**.

DNA and RNA: Structure & Function

What is DNA?



DNA (Deoxyribonucleic Acid) is the **genetic material** in most organisms, responsible for storing and transmitting hereditary(বংশগত) information.

What is RNA?



RNA (Ribonucleic Acid) plays a crucial role in protein synthesis and gene expression, acting as a messenger between DNA and ribosomes.

Structure of DNA

1. DNA Composition

DNA is a **double-stranded** molecule made of **nucleotides**, which consist of:

- a. **Deoxyribose sugar** (5-carbon sugar)
- b. **Phosphate group**
- c. **Nitrogenous base** (Adenine, Thymine, Cytosine, Guanine)

2. DNA Structure

- a. **Double helix shape** (discovered by Watson & Crick)
- b. **Antiparallel strands** (one strand runs 5' → 3', the other runs 3' → 5')
- c. **Base pairing rule (Chargaff's Rule):**
 - i. **Adenine (A) pairs with Thymine (T)** (2 hydrogen bonds)
 - ii. **Cytosine (C) pairs with Guanine (G)** (3 hydrogen bonds)

3. Function of DNA

- a. Stores genetic information.
- b. Guides **protein synthesis** through transcription and translation.
- c. Transfers hereditary traits to offspring.

Structure of RNA

1. RNA Composition

RNA is a **single-stranded** molecule made of:

- a. **Ribose sugar** (instead of deoxyribose)
- b. **Phosphate group**
- c. **Nitrogenous bases** (Adenine, **Uracil**, Cytosine, Guanine)

2. RNA Structure & Types

Unlike DNA, RNA has **three major types** with different functions:

Type of RNA	Function
mRNA (Messenger RNA)	Carries genetic instructions from DNA to ribosomes.
tRNA (Transfer RNA)	Brings amino acids to ribosomes for protein synthesis.
rRNA (Ribosomal RNA)	Forms ribosomes, which assemble proteins.

3. Function of RNA

- mRNA** transcribes genetic code from DNA and takes it to ribosomes.
- tRNA** helps decode the mRNA sequence into proteins.
- rRNA** forms the core of ribosomes and catalyzes protein synthesis.

Key Differences Between DNA & RNA

Feature	DNA	RNA
Strands	Double-stranded	Single-stranded
Sugar	Deoxyribose	Ribose
Nitrogen Bases	A, T, C, G	A, U, C, G
Stability	More stable	Less stable
Location	Nucleus (mostly)	Cytoplasm & nucleus
Function	Stores genetic information	Protein synthesis & gene regulation

Phosphate Group: Definition & Function

Definition



A **phosphate group** is a chemical group consisting of **one phosphorus (P) atom** bonded to **four oxygen (O) atoms** (PO_4^{3-}). It is a crucial component of many biological molecules, including **DNA, RNA, ATP, and phospholipids**.

Structure of a Phosphate Group

- Central **phosphorus (P) atom**.
- Four **oxygen (O) atoms** bonded to phosphorus.
- Carries a **negative charge (-3 overall charge in free form)**.
- Often involved in **high-energy bonds**.

Function of the Phosphate Group in Biology

Molecule	Role of Phosphate Group
DNA & RNA	Forms the sugar-phosphate backbone , linking nucleotides via phosphodiester bonds .
ATP (Adenosine Triphosphate)	Stores and releases energy by breaking phosphate bonds (ATP → ADP + P).
Phospholipids (Cell Membranes)	Forms the hydrophilic head of phospholipids, making up the cell membrane bilayer .
Proteins (Phosphorylation)	Regulates protein activity in cell signaling and metabolism.

Phosphate Group in DNA & RNA

- Links **nucleotides** together through **phosphodiester bonds**.
- Provides DNA and RNA with a **negative charge**, making them **soluble in water**.
- Essential for **genetic information storage and transfer**.

Nitrogenous Base: Definition & Function



A **nitrogenous base** is a nitrogen-containing molecule that plays a key role in **genetic coding** by forming the fundamental units of **DNA and RNA**. These bases pair with each other to store and transmit genetic information.

Types of Nitrogenous Bases

There are **five nitrogenous bases**, classified into two groups:

1. **Purines (Double-Ring Structure)**
 - a. **Adenine (A)**
 - b. **Guanine (G)**
2. **Pyrimidines (Single-Ring Structure)**
 - a. **Cytosine (C)**
 - b. **Thymine (T)** (Only in DNA)

c. **Uracil (U)** (Only in RNA, replaces Thymine)

Nitrogenous Bases in DNA and RNA

Base	Found in DNA?	Found in RNA?	Base Pairing
Adenine (A)	✓	✓	Pairs with Thymine (T) in DNA, Uracil (U) in RNA
Thymine (T)	✓	✗	Pairs with Adenine (A) in DNA
Uracil (U)	✗	✓	Pairs with Adenine (A) in RNA
Guanine (G)	✓	✓	Pairs with Cytosine (C)
Cytosine (C)	✓	✓	Pairs with Guanine (G)

Function of Nitrogenous Bases

1. **Store Genetic Information** – The sequence of bases encodes genetic instructions.
2. **Base Pairing** – Forms **hydrogen bonds** to create the double-helix structure.
3. **Facilitates Transcription & Translation** – Determines the sequence of mRNA and protein synthesis.

Base Pairing Rules (Chargaff's Rule)

- **A pairs with T** (or U in RNA) → **2 hydrogen bonds**
- **G pairs with C** → **3 hydrogen bonds** (stronger bond)

Deoxyribose Sugar: Definition & Function

Definition



Deoxyribose is a **five-carbon (pentose) sugar** that forms the structural backbone of **DNA (Deoxyribonucleic Acid)**. It is responsible for linking **phosphate groups** and **nitrogenous bases** to create the **DNA double helix**.

Structure of Deoxyribose

- **Chemical Formula:** $C_5H_{10}O_4$
- **Pentose (5-carbon) sugar** with a **ring structure**.
- **Lacks an oxygen atom** at the **2' carbon** (compared to ribose in RNA).

Key Differences Between Deoxyribose and Ribose

Feature	Deoxyribose (DNA Sugar)	Ribose (RNA Sugar)
Chemical Formula	$C_5H_{10}O_4$	$C_5H_{10}O_5$
Oxygen at 2' Carbon?	No (H instead of OH)	Yes (OH group present)
Stability	More stable	Less stable
Found in	DNA	RNA

Functions of Deoxyribose in DNA

1. Forms the Sugar-Phosphate Backbone

- a. Bonds with **phosphate groups** via **phosphodiester bonds** to create the DNA strand.

2. Supports Base Pairing

- a. **Attaches to nitrogenous bases** (A, T, C, G) at the **1' carbon**.

3. Increases DNA Stability

- a. The **lack of oxygen at the 2' carbon** makes DNA **chemically more stable** than RNA.

Why Is Deoxyribose Important?

- Helps maintain the **double-helix structure** of DNA.
- Ensures **genetic information storage** and **accurate replication**.
- Increases DNA's **resistance to degradation**, making it ideal for long-term genetic storage.

Ribose Sugar: Definition & Structure

Definition



Ribose is a **five-carbon (pentose) sugar** that forms the backbone of **RNA (Ribonucleic Acid)** and is also a key component of **ATP (Adenosine Triphosphate)**, **NADH**, and **other biomolecules**.

Structure of Ribose

- **Chemical Formula:** $C_5H_{10}O_5$
- **Pentose (5-carbon) sugar** with a **ring structure**.
- **Contains an -OH (hydroxyl) group** at the **2' carbon**, making it more reactive than deoxyribose.

Difference Between Ribose and Deoxyribose

Feature	Ribose (RNA Sugar)	Deoxyribose (DNA Sugar)
Chemical Formula	$C_5H_{10}O_5$	$C_5H_{10}O_4$
2' Carbon Group	OH (Hydroxyl group present)	H (No oxygen, "deoxy")
Function	Found in RNA, ATP, NADH	Found in DNA
Stability	Less stable (due to OH group)	More stable

Functions of Ribose Sugar

1. **Forms the Sugar-Phosphate Backbone**
 - a. Attaches to **phosphate groups** via **phosphodiester bonds** in RNA.
2. **Supports Genetic Coding in RNA**
 - a. Bonds with **nitrogenous bases (A, U, C, G)** to store genetic instructions.
3. **Key Component in ATP**
 - a. Provides **energy** in cells through **ATP (Adenosine Triphosphate)**.
4. **Involved in Metabolism**
 - a. Found in **NADH, FADH₂, and coenzymes** essential for energy production.

Why Is Ribose Important?

- Essential for **protein synthesis** in RNA.
- Drives **cellular energy transfer** in ATP.
- Supports **metabolic reactions** in living organisms.

Chromosome: Definition & Structure

Definition



A **chromosome** is a thread-like structure made of **DNA and proteins (histones)** that carries **genetic information** in the form of **genes**. Chromosomes are found in the **nucleus** of eukaryotic cells and in the cytoplasm of prokaryotic cells.

Structure of a Chromosome

1. **DNA Double Helix** – The basic genetic material.
2. **Nucleosome** – DNA wraps around **histone proteins** to form nucleosomes.
3. **Chromatin Fiber** – Nucleosomes coil into a dense fiber.
4. **Chromatid** – A condensed, organized structure of chromatin.
5. **Centromere** – The central region where **sister chromatids** attach.
6. **Telomeres** – Protective ends that prevent chromosome degradation.

Types of Chromosomes

1. **Autosomes (Body Chromosomes)** – Control general body traits. (Humans have **22 pairs**.)
2. **Sex Chromosomes** – Determine gender (**XX for females, XY for males** in humans).

Functions of Chromosomes

- **Store Genetic Information** – Contain DNA that codes for proteins.

- **Ensure Accurate Cell Division** – Prevents genetic errors during **mitosis and meiosis**.
- **Regulate Gene Expression** – Chromatin modification controls gene activity.
- **Maintain Genetic Stability** – Telomeres protect DNA from damage.

Gene and Gene Regulation

Definition



A **gene** is a segment of **DNA** that contains the instructions for making a **specific protein** or **RNA molecule**. Genes are the **functional units of heredity** and determine an organism's traits by coding for proteins that regulate biological functions.

Structure of a Gene

A gene consists of:

1. **Promoter** – The starting point where RNA polymerase binds to initiate transcription.
2. **Coding Region** – The part of the gene that contains the sequence for making proteins.
3. **Terminator** – The end sequence that signals transcription to stop.

What is Gene Regulation?



Gene regulation is the process by which cells control which genes are **turned on or off** at different times. This allows cells to respond to changes in the environment and perform specialized functions.

Mechanisms of Gene Regulation

1. **Transcriptional Regulation (Control before mRNA is made)**

- a. **Promoter & Enhancers** – Help RNA polymerase bind and start transcription.
 - b. **Transcription Factors** – Proteins that turn genes **on** or **off**.
 - c. **Epigenetic Modifications** – DNA methylation and histone modification affect gene activity.
2. **Post-Transcriptional Regulation (Control after mRNA is made)**
 - a. **RNA Splicing** – Removes introns (non-coding regions) and joins exons (coding regions).
 - b. **RNA Interference (RNAi)** – Small RNAs (miRNA, siRNA) can block translation of specific mRNAs.
3. **Translational Regulation (Control during protein synthesis)**
 - a. **Ribosome Activity** – Proteins or RNA structures can block ribosome binding to mRNA.
4. **Post-Translational Regulation (Control after protein is made)**
 - a. **Protein Modifications** – Phosphorylation, acetylation, or degradation affect protein function.
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Why is Gene Regulation Important?

- **Cell Differentiation** – Ensures that different cells (e.g., muscle, nerve) express only necessary genes.
- **Response to Environment** – Helps adapt to changes (e.g., producing insulin after eating).
- **Prevention of Diseases** – Errors in gene regulation can cause cancer and genetic disorders.

DNA Template Strand vs. Coding Strand

In **DNA transcription**, two strands play different roles:

Template Strand (Antisense Strand)

- Acts as a **blueprint** for mRNA synthesis.

- Runs in the **3' → 5'** direction.
- Complementary to the mRNA sequence (**except U replaces T**).
- Example: **3' - TAC GGT ACC - 5'**
- Produces mRNA: **5' - AUG CCA UGG - 3'**

Coding Strand (Sense Strand)

- Same sequence as mRNA (**except T instead of U**).
- Runs in the **5' → 3'** direction.
- Does **not** participate in transcription directly.
- Example: **5' - ATG CCA TGG - 3'**

Key Difference:

- **Template strand** is used by **RNA polymerase** to make mRNA.
- **Coding strand** is identical to mRNA (except T → U).